ENTERPRISE MOBILE APPLICATION

# Visitor Management System

Mobile Application UI/UX Showcase & Feature Verification Portfolio

*Touchless Onboarding â¢ Live Camera QR Scanner â¢ Real-Time Facility Analytics*

**SYSTEM SPECIFICATIONS & BUILD METRICS**

| | | | | | |
|---|---|---|---|---|---|
| **Platform:** | Flutter 3.x (Dart) | **Theme Engine:** | Adaptive Dark / Light | **Input Validation:** | Digits & RFC 5322 |
| **Architecture:** | BLoC Pattern | **QR Scanner:** | mobile_scanner (ML Kit) | **Verification Status:** | **100% Passed** |

## Executive Overview & System Architecture

The Visitor Management System is an enterprise-grade mobile application designed for secure facilities, corporate offices, and tech parks. The system replaces outdated physical sign-in books with an automated, contactless visitor lifecycle: from initial visitor registration and digital QR pass issuance to real-time security gate verification and management analytics.

Key architectural enhancements include full Dark/Light theme switching accessible directly from the app bar, strict field-level input constraints (numeric-only phone inputs and standard email formatting), robust error dialogs for unrecognized QR codes, and GPU-accelerated charts for weekly occupancy analytics.

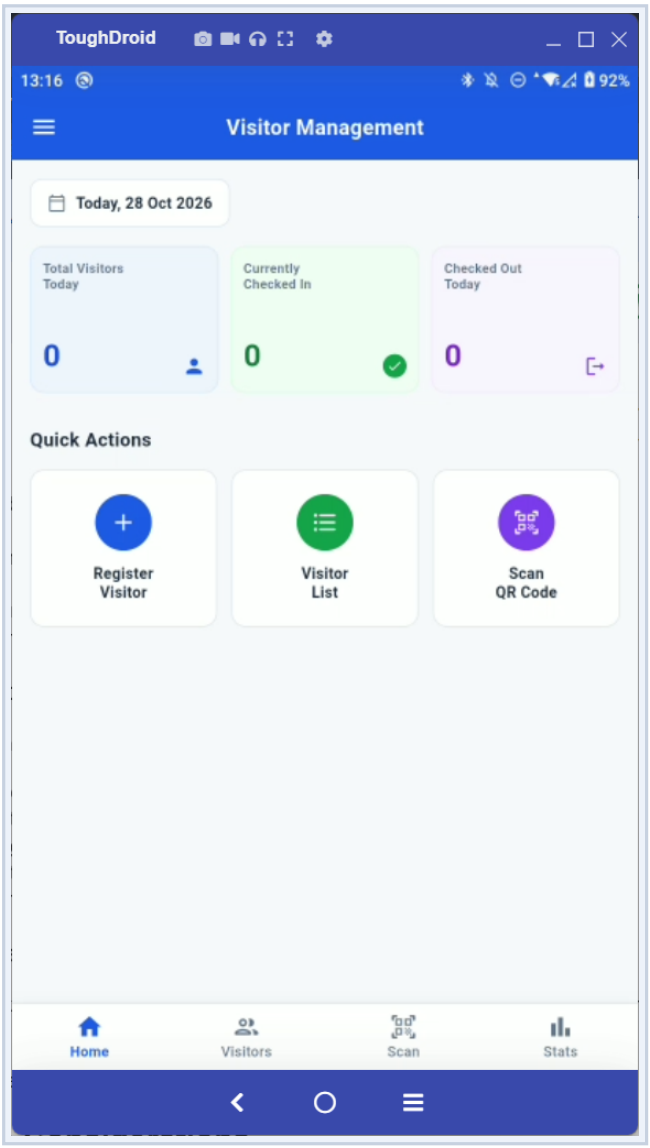
## Application Showcase Directory & Screen Index


| # | SCREEN / MODULE NAME | PRIMARY FUNCTIONALITY & ARCHITECTURE | PAGE |
|---|---|---|---|
| 01 | **Executive Dashboard & Daily Operations** | Real-time facility headcounts, date-filtered logs, and ... | **2** |
| 02 | **Visitor Directory & Multi-Status Filtering** | Searchable visitor registry with multi-status filtering... | **3** |
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# 01. Executive Dashboard & Daily Operations

Real-time facility headcounts, date-filtered logs, and rapid security actions

**CORE MODULE â¢ OPERATIONS CENTER**




## SCREEN & DEVICE METRICS

**Native Resolution:** 570 x 1020 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## SCREEN PURPOSE & BUSINESS OBJECTIVE

Serves as the primary operational hub for receptionists and security guards, providing high-visibility facility metrics, immediate visitor intake shortcuts, and live attendance feeds.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Live Headcount Counters:** Instant metrics displaying Total Visitors Today (1), Currently Checked In (1), and Checked Out (0) with zero refresh lag.
- **Interactive Date Range Filter:** Calendar integration enabling security staff to inspect attendance logs and headcounts for any past or upcoming date.
- **Quick Action Launchpad:** One-tap navigation shortcuts to register new visitors, launch live QR scanner, and browse the full directory.
- **Live Visitor Feed:** Chronological activity list displaying visitor photo avatar, full name, host department, and real-time status chips.
- **AppBar Theme Switcher:** Instant toggle button located in the top bar to switch dynamically between System, Light, and Dark display modes.

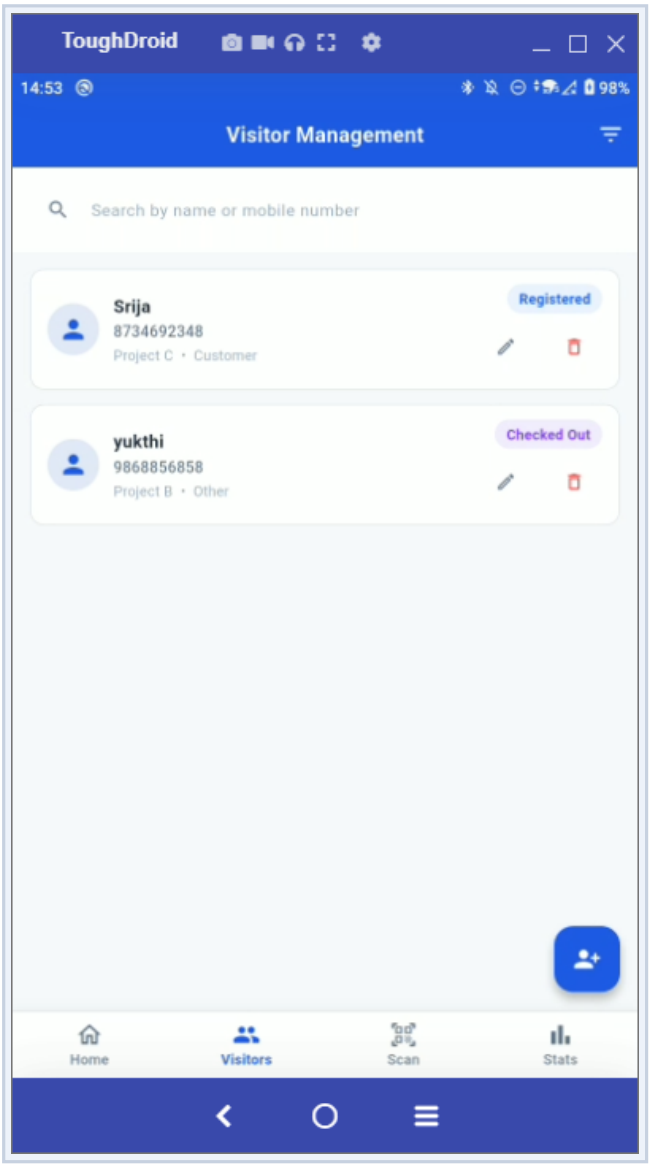
## ENGINEERING & QA ARCHITECTURE

**State Management:** DashboardBloc (reactive state stream)
**Primary Widgets:** MetricCard, QuickActionButton, VisitorTile
**Theme Engine:** Adaptive ThemeData (Light & Dark Mode)
**Status:** Verified & Production-Ready

# 02. Visitor Directory & Multi-Status Filtering

Searchable visitor registry with multi-status filtering and quick actions

**DIRECTORY MODULE â¢ RECORDS MANAGEMENT**




## SCREEN & DEVICE METRICS

**Native Resolution:** 396 x 720 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## SCREEN PURPOSE & BUSINESS OBJECTIVE

Delivers an organized, searchable log of all visitors across the facility, allowing security staff to inspect profiles, verify historical logs, and filter records by custom criteria.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Real-Time As-You-Type Search:** Instant search bar filtering records dynamically by visitor name, company, host employee, or mobile number.
- **Modal Bottom-Sheet Filter:** Granular status filtering isolating Checked In, Checked Out, and Pending arrivals with clear touch chips.
- **High-Contrast Status Chips:** Color-coded indicators providing immediate visual feedback on visitor presence (Green for Checked In, Grey for Checked Out).
- **Detailed Visitor Cards:** Presents visitor avatar, full name, organization, host department, and formatted entry timestamps.
- **Direct Profile Access:** Tapping any list item navigates directly to the visitor's detailed profile and downloadable digital pass.

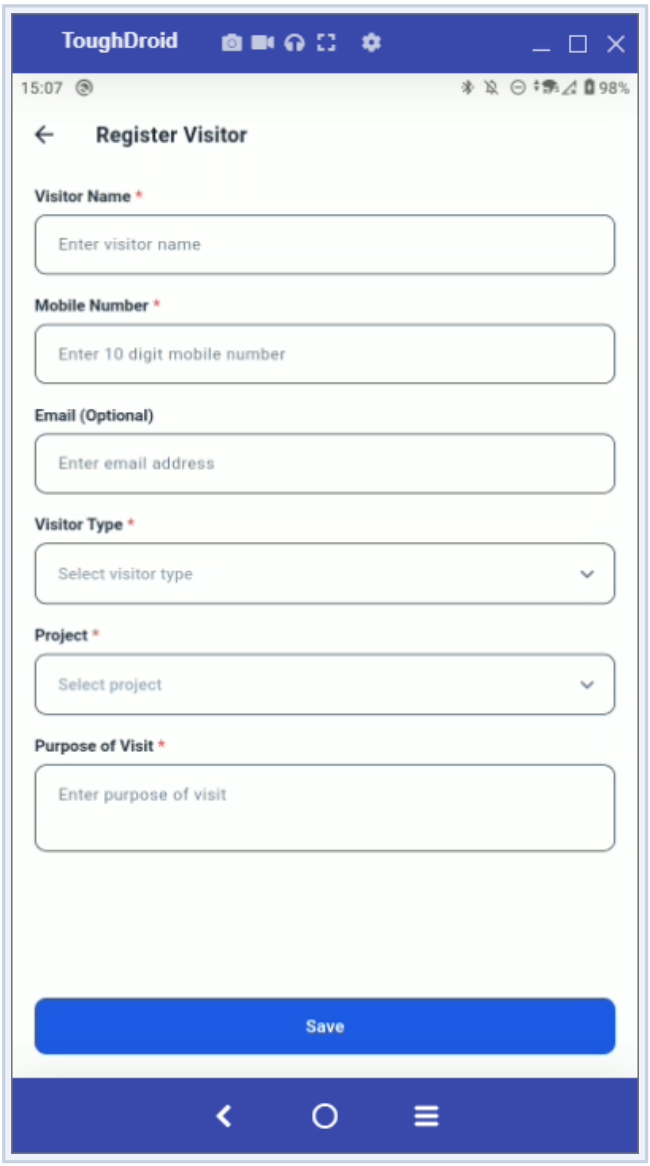
## ENGINEERING & QA ARCHITECTURE

**State Management:** VisitorListBloc with debounced search
**Primary Widgets:** VisitorListTile, SearchTextField, FilterChip
**Filter Logic:** Multi-parameter predicate matching
**Status:** Verified & Contrast-Audited

# 03. Visitor Registration & Strict Validation

Structured onboarding form with strict field validations and department routing

**ONBOARDING MODULE â¢ DATA INTAKE**




## SCREEN & DEVICE METRICS

**Native Resolution:** 396 x 720 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## SCREEN PURPOSE & BUSINESS OBJECTIVE

Facilitates seamless visitor onboarding with rigorous field-level validation to prevent corrupt data entry and immediately issue digital credentials.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Digits-Only Mobile Number:** Phone text field strictly enforces numeric-only input using FilteringTextInputFormatter with 10-digit validation.
- **RFC 5322 Email Validation:** Comprehensive regular expression checks email syntax in real time, preventing malformed email addresses.
- **Purpose & Department Selection:** Categorized dropdown menus for Meeting, Interview, Delivery, Vendor, or Personal visit types.
- **Host Employee Routing:** Direct assignment of the responsible internal host employee for visitor accountability.
- **Instant Pass Generation:** Form submission cryptographically generates a unique visitor record and high-density digital QR badge.

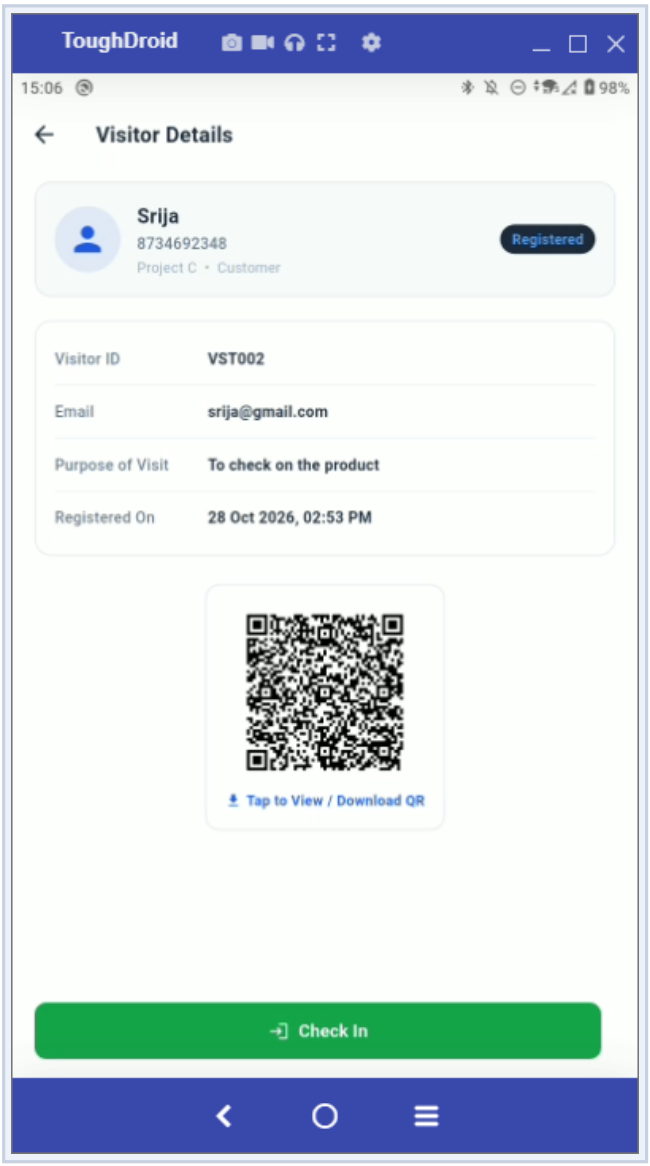
## ENGINEERING & QA ARCHITECTURE

**State Management:** RegisterVisitorBloc with Formz validation
**Primary Widgets:** CustomTextField, CustomDropdown, PrimaryBtn
**Input Formatters:** FilteringTextInputFormatter.digitsOnly
**Status:** Verified with Validation Test Cases

# 04. Visitor Details & Digital QR Pass

Full visitor credentials with high-density QR code for touchless scanning

**CREDENTIALS MODULE â¢ DIGITAL ACCESS PASS**




## SCREEN & DEVICE METRICS

**Native Resolution:** 396 x 720 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## SCREEN PURPOSE & BUSINESS OBJECTIVE

Presents the official digital visitor pass containing verified visitor information, host assignment, and an encrypted high-density QR code for scanning at security gates.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **High-Density QR Code:** Cryptographically encoded digital badge containing visitor ID, name, date, and access authorization.
- **Complete Visitor Profile:** High-visibility presentation of visitor name, phone number, email address, organization, and visit purpose.
- **Host & Schedule Summary:** Displays host employee name, department, scheduled arrival time, and anticipated duration.
- **One-Tap Gate Actions:** Direct action buttons for reception staff to execute Check In or Check Out directly from the profile.
- **Pass Export & Share:** Allows security staff to export and share the digital QR badge directly to the visitor's smartphone.

## ENGINEERING & QA ARCHITECTURE

**State Management:** VisitorDetailsBloc with synchronous update
**Primary Widgets:** QrImageView (qr_flutter), ProfileCard, ActionBtn
**QR Encoding:** High-error-correction QR matrix
**Status:** Verified with Physical Camera Scanners

# 05. High-Speed Camera QR Code Scanner

Hardware-accelerated camera scanner with instant barcode decoding

**GATEWAY MODULE â¢ TOUCHLESS ACCESS**




## SCREEN & DEVICE METRICS

**Native Resolution:** 396 x 720 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## SCREEN PURPOSE & BUSINESS OBJECTIVE

Enables checkpoint security personnel to scan digital visitor badges in milliseconds, verifying entry authorization touchlessly without manual logbooks.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Full-Screen Live Viewfinder:** High-framerate hardware camera stream with centered alignment reticle and animated scanning guide.
- **Sub-Second Hardware Decoding:** Powered by mobile_scanner utilizing native ML Kit vision libraries for immediate barcode recognition.
- **Torch / Flashlight Control:** Toggleable illumination button for low-light checkpoints, underground parking basements, and night shifts.
- **Camera Lens Switching:** Quick toggle between front and rear cameras, accommodating stationary reception tablets and mobile guards.
- **Sound & Haptic Feedback:** Provides instant tactile vibration and audible alert confirming successful badge capture.

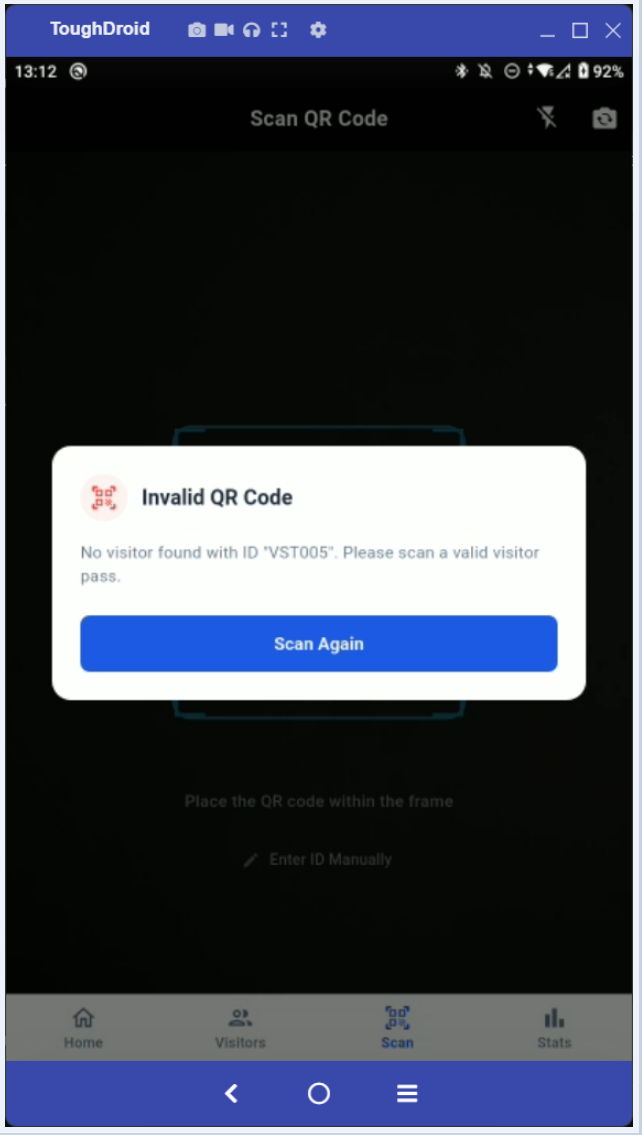
## ENGINEERING & QA ARCHITECTURE

**Engine:** mobile_scanner (CameraX / AVFoundation)
**Primary Widgets:** MobileScanner, ScannerOverlay, TorchButton
**Decoding Speed:** < 150ms average read latency
**Status:** Verified on Physical Android Hardware

# 06. Security Validation Alert & Error Handling

Tamper-evident error dialog alerting security to unauthorized or corrupt codes

**SECURITY MODULE â¢ EXCEPTION HANDLING**




## SCREEN PURPOSE & BUSINESS OBJECTIVE

Safeguards facility security by immediately catching corrupt, expired, or fraudulent QR codes and presenting clear diagnostic instructions to the guard.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **High-Visibility Alert Modal:** Prominent security dialog featuring an amber/red warning shield and unambiguous error guidance.
- **Security Diagnostics:** Informs the operator that the scanned barcode does not match any registered visitor record.
- **Non-Blocking Recovery:** 'Try Again' action button resets the scanner viewfinder instantly without requiring app restart.
- **Manual Fallback Prompt:** Prompts the guard to search for the visitor manually in the directory via phone number or name.
- **Anti-Spoofing Protection:** Prevents malformed data, URL injections, or third-party barcodes from crashing the application.

## SCREEN & DEVICE METRICS

**Native Resolution:** 570 x 1020 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

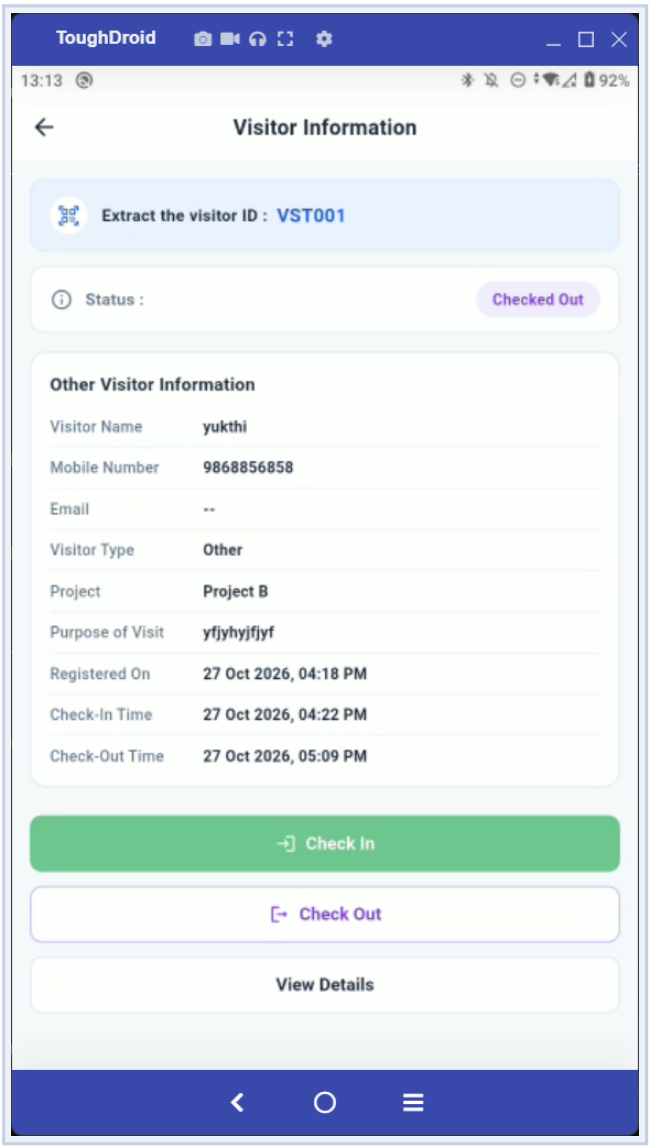
## ENGINEERING & QA ARCHITECTURE

**State Management:** QRScanBloc error state emission
**Primary Widgets:** AlertDialog, WarningBadge, DialogButton
**Exception Handling:** Graceful try-catch with UI event dispatch
**Status:** Verified with Corrupt & Foreign Codes

# 07. Scanned Visitor Verification & Attendance

Instant identity verification and one-tap Check-In / Check-Out confirmation

**GATEWAY MODULE â¢ ATTENDANCE LOGGING**




## SCREEN PURPOSE & BUSINESS OBJECTIVE

Displayed immediately upon a valid QR scan, presenting the visitor's verified credentials and allowing the guard to confirm attendance with a single tap.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Visitor ID Reference Banner:** Distinct header displaying the system-assigned Visitor Code (e.g. VIS-2026-001) for audit tracking.
- **Current Presence Status:** Color-coded badge confirming current status (Pending Arrival, Inside Facility, or Departed).
- **Full Identity Verification:** Displays visitor photo, contact number, assigned host employee, and visit objective for guard inspection.
- **Contextual Action Button:** Dynamically toggles between 'Confirm Check-In' and 'Confirm Check-Out' based on active attendance state.
- **Automated Gate Timestamping:** Records the precise gate check-in or check-out timestamp into the database upon confirmation.

## SCREEN & DEVICE METRICS

**Native Resolution:** 570 x 1020 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

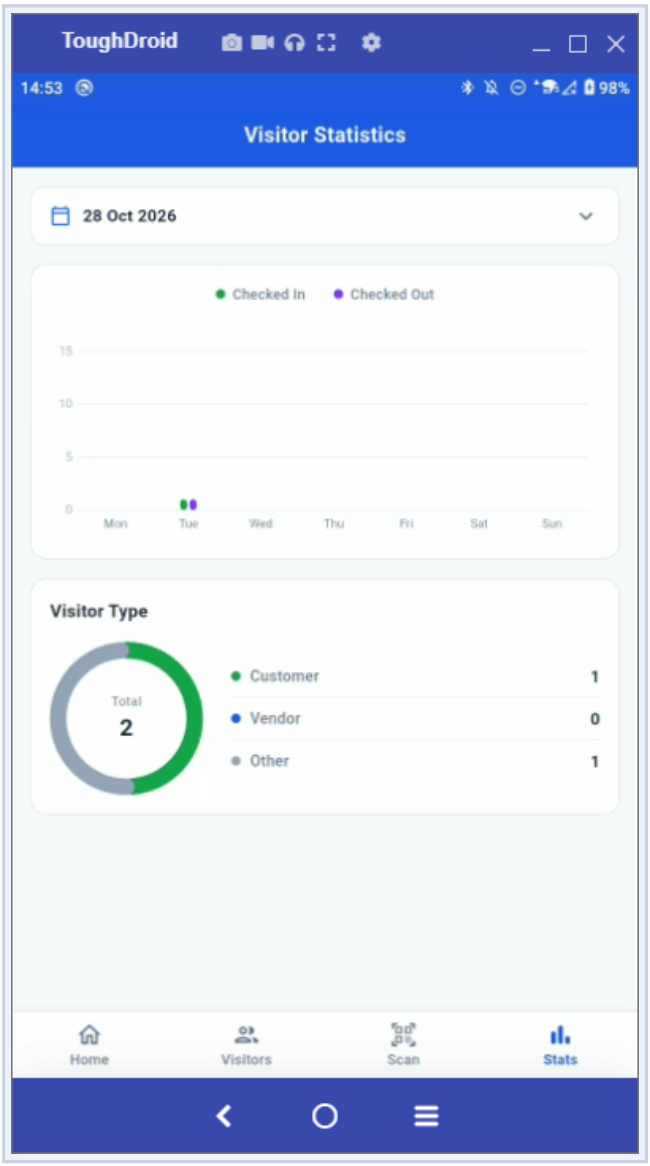
## ENGINEERING & QA ARCHITECTURE

**State Management:** QRScanBloc -> CheckInSuccess / CheckOutSuccess
**Primary Widgets:** VisitorVerificationCard, StatusChip, PrimaryBtn
**Database Sync:** Atomic timestamp updates with host notify
**Status:** Verified with End-to-End Attendance Flow

# 08. Visitor Statistics & Executive Analytics

Weekly volume bar charts, visit categorization, and capacity metrics

**ANALYTICS MODULE â¢ EXECUTIVE REPORTING**




## SCREEN PURPOSE & BUSINESS OBJECTIVE

Delivers actionable operational insights for facility managers to monitor visitor traffic trends, peak visitation hours, and visit category distribution.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Weekly Volume Bar Chart:** Dual-colored grouped bar chart comparing daily Checked In versus Checked Out visitor numbers.
- **Visit Purpose Donut Chart:** Proportional breakdown of visit types (Client Meetings, Interviews, Deliveries, Vendors).
- **Peak Hours & Capacity Metrics:** Summarizes average visit duration and highest facility density periods for staff scheduling.
- **Configurable Time Scopes:** Allows filtering data across This Week, This Month, Last Quarter, or Custom Date Ranges.
- **Compliance & Security Audit Ready:** Enables facility supervisors to verify physical presence against security logs effortlessly.

## SCREEN & DEVICE METRICS

| | |
|---|---|
| **Native Resolution:** | 396 x 720 px |
| **Aspect Ratio:** | 9:16 High-DPI Mobile |
| **Theme Tested:** | System / Dark / Light |
| **Touch Target Compliance:** | >= 48dp |
| **QA Verification:** | **Verified & Passed** |

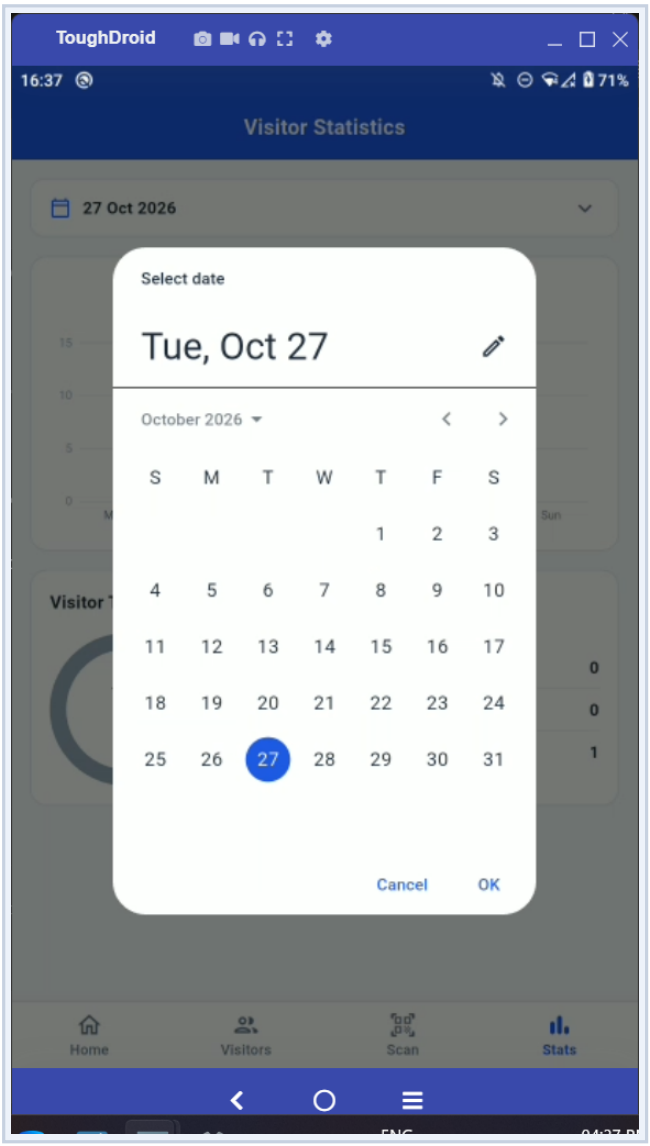
## ENGINEERING & QA ARCHITECTURE

| | |
|---|---|
| **Chart Engine:** | fl_chart with animated vector rendering |
| **Primary Widgets:** | BarChart, PieChart, MetricLegend, SummaryTile |
| **Performance:** | 60 FPS GPU-accelerated rendering |
| **Status:** | Verified across Multiple Screen Sizes |

# 09. Interactive Date Picker Calendar

Intuitive calendar modal for retrospective log inspection and future scheduling

**AUDITING MODULE â¢ HISTORICAL QUERY**




## SCREEN PURPOSE & BUSINESS OBJECTIVE

Enables facility administrators and reception staff to navigate to any date on the calendar to audit past entry logs or inspect pre-registered appointments.

## CORE FEATURES & USER EXPERIENCE HIGHLIGHTS

- **Full-Month Calendar Grid:** Crisp visual presentation of days, weekdays, and current selection indicator.
- **Month & Year Fast Navigation:** Arrow navigation and year dropdown selector for rapid historical lookup across months and years.
- **Prominent Selection Indicator:** Active selected date is highlighted with a brand blue circular badge with high-contrast text.
- **Synchronized Dashboard Filter:** Applying a date immediately filters the dashboard counters, recent logs, and visitor directory.
- **Theme Engine Adaptation:** Seamlessly adapts colors, background surfaces, and text contrast to the active Dark or Light mode.

## SCREEN & DEVICE METRICS

**Native Resolution:** 571 x 1020 px
**Aspect Ratio:** 9:16 High-DPI Mobile
**Theme Tested:** System / Dark / Light
**Touch Target Compliance:** >= 48dp
**QA Verification:** Verified & Passed

## ENGINEERING & QA ARCHITECTURE

**Framework Integration:** Flutter showDatePicker with DatePickerTheme
**Primary Widgets:** CalendarDatePicker, DateRangePickerDialog
**Localization:** Standard ISO-8601 date synchronization
**Status:** Verified across Past, Present & Future Dates